NON-MONOTONICITY IN k AT $n = 3$: A CENSUS CONTRADICTING A k-MONOTONICITY CLAUSE TRANSCRIBED FROM D'ANGELI AND RODARO

Abstract. For a uniformly random k-out digraph G on n labelled vertices — each vertex choosing its k out-edge heads independently and uniformly *with replacement* — let $P(n,k) = \Pr\big(G \text{ totally simple} \mid G \text{ strongly connected}\big)$. A three-clause conjecture of D'Angeli and Rodaro, as we transcribe it by hand from the e-print (Section 1), asserts among other things that for each fixed n the map $k \mapsto P(n,k)$ is non-decreasing on $k \geq 2$. An exhaustive census at $n = 3$ contradicts that clause. Writing $B(n,k)$ for the total weight of the strongly connected states and $A(n,k)$ for the total weight of those that are also totally simple,

$$A(3,2) = 114, \quad B(3,2) = 296, \qquad A(3,3) = 5132, \quad B(3,3) = 13754,$$

so $P(3,2) = \frac{57}{148} > \frac{2566}{6877} = P(3,3)$, the inequality being decided by $114 \cdot 13754 = 1\,567\,956 > 1\,519\,072 = 5132 \cdot 296$. All four integers have closed forms in k that a reader can evaluate by hand, so no computation is needed to check the inequality. Two caveats bound what this settles. The clause, its quantifiers and the probability model are transcribed by hand from the cited e-print, of which we can supply neither an immutable copy nor a hash; and the drop, about 1.2 percentage points, disappears under the rival uniform measure on labelled k-out multigraphs, so Section 2 records what pins the with-replacement reading. Nothing here bears on the two *asymptotic* clauses of the same conjecture, which remain open. The denominators are not new: $B(n,2)$ is a published 1965/1971 automaton census, and $B(3,3) = 13754$ is on record as well.

1. The clause, as transcribed, and its locator

The source is D. D'Angeli and E. Rodaro, *On totally synchronizing graphs*, arXiv:2607.17335 [1], version v1, submitted 19 July 2026. All line references are to its LaTeX e-print source rather than to the compiled PDF: the file `totally_main.tex`, 143 403 bytes, 1977 lines. In the section "Totally simple graphs" the source states a three-clause conjecture, tex lines 1457–1471; it is unnumbered in the source, carries no `\cite`, and is attributed to nobody there.

We can supply neither an immutable copy of that e-print nor a cryptographic hash of it, and we have not confirmed that the arXiv identifier resolves for a third party. Everything this note attributes to the source —

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the clause quoted below, its ambient quantifiers, the definitions of Section 2 and the probability model — is hand transcription from the file named above, and it is not checked by the accompanying program (Section 6). A reader who cannot obtain the file cannot check that part. What is proved here without reference to the source is the arithmetic of Theorem 1: $P(3,2) > P(3,3)$ in the model of Section 2.

The clause at issue is the second one, tex line 1463, transcribed verbatim:

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Moreover, for fixed $n$ the function $k \longmapsto
\Pr(\text{$G$ is totally simple}\mid \mathrm{SCC})$
is nondecreasing.
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The conjecture opens "Fix $k \geq 2$", so on this transcription the clause asserts

$$P(n,k) \;\leq\; P(n,k+1) \qquad \text{for every } n \geq 1 \text{ and every } k \geq 2, \tag{1}$$

where $P(n,k) = \Pr(G \text{ totally simple} \mid \mathrm{SCC})$. *The clause places no lower bound on n.* The first clause of the same conjecture (tex line 1461) asserts $P(n,k) \to 1$ as $n \to \infty$ for fixed k, and the third (tex lines 1465–1470) asserts an expansion $P(n,k) = 1 - \exp(-c_k n + \beta_k) + o(1)$ with c_k increasing in k. Both of those are asymptotic; nothing below bears on them.

2. The model and the predicate

The measure. The source fixes the model in the paragraph at tex lines 1445–1455: "in our model each vertex chooses independently k outgoing edges with replacement, and the head of each edge is uniform in V". So, with $V = \{0, \ldots, n-1\}$, the probability space is V^{kn} with the uniform measure. Only the out-multiset of each vertex matters for what follows, so we work with the *state* $c = (c_{vw})_{v,w \in V}$, where c_{vw} is the number of edges $v \to w$ and $\sum_w c_{vw} = k$ for every v; loops and parallel edges are allowed. The number of ordered choices realising a state is its weight

$$W(c) = \prod_{v \in V} \frac{k!}{\prod_{w \in V} c_{vw}!}, \qquad \text{and} \qquad \sum_c W(c) = n^{kn}. \tag{2}$$

Write $a_{vw} := c_{vw}$ when the matrix notation is more convenient.

The predicate. For a partition $\mathcal{C}$ of V the source calls $\mathcal{C}$ *lumpable* (tex 339–343) if for every pair of blocks C_i, C_j and all $u, u' \in C_i$ one has $|\delta_u(C_j)| = |\delta_{u'}(C_j)|$, where $\delta_u(C_j)$ is the set of edges from u into C_j. A congruence different from the discrete relation Δ_G and the universal relation ∇_G is called *nontrivial* (tex 355), and an automaton is *simple* when its only congruences are the discrete and the universal one (tex 240). Chaining that with the definition of *totally simple* (tex 1357–1359), with the source's lemma identifying lumpable partitions with congruences (tex 1375–1377) and with its theorem relating lumpability to simplicity (tex 1386–1388) — and with the source's own restatement at tex 1510, "total simplicity is equivalent to the absence of nontrivial lumpable partitions" — gives the form we use:

$$G \text{ is } \textit{totally simple} \iff \text{no partition of } V \text{ other than } \Delta_G \text{ and } \nabla_G \text{ is lumpable.} \tag{3}$$

Both Δ_G and ∇_G are always lumpable (∇_G because $|\delta_u(V)| = k$ for every u; Δ_G vacuously), so excluding both is forced by the source rather than chosen by us, and the predicate is well posed.

The three quantities. Set

$$B(n,k) = \sum_{c \text{ str. conn.}} W(c), \qquad A(n,k) = \sum_{\substack{c \text{ str. conn.} \\ c \text{ totally simple}}} W(c), \tag{4}$$

$$P(n,k) = \frac{A(n,k)}{B(n,k)}. \tag{5}$$

Then $P(n,k) = \Pr(G \text{ totally simple} \mid \text{SCC})$ in the source's model.

Remark (the model is load-bearing). "Uniformly random k-out graph" also admits a rival reading: the uniform measure on labelled k-out multigraphs, i.e. weight 1 per state instead of (2). Under that rival reading the counterexample below *does not exist*: writing P_u for the rival quantity, $P_u(3,2) = \frac{34}{65} < \frac{133}{243} = P_u(3,3)$, by $34 \cdot 486 = 16\,524 < 17\,290 = 266 \cdot 65$ on the unreduced rival state counts, and every rival k-step we computed is nondecreasing. Since the drop found below is only about 1.2 percentage points, the choice of model is the difference between a counterexample and none. Two features of the source pin the ordered reading used here. First, tex 1449 says "with replacement" in words. Second, the source's own displayed identity at tex 1452, $\Pr(\delta_V(v) = \emptyset) = (1 - 1/n)^{kn}$ for the event that no edge lands on v, is true in the ordered model and false in the rival one: at $n = 3, k = 2$ it gives $(2/3)^6 = 64/729$, whereas the rival measure gives $(3/6)^3 = 1/8$. The residual risk in this note is therefore a *reading* risk, not an arithmetic one.

3. The counterexample at $n = 3$

Theorem 1. *In the model of Section 2,*

$$A(3,2) = 114, \qquad B(3,2) = 296, \qquad A(3,3) = 5132, \qquad B(3,3) = 13754,$$

so that

$$P(3,2) = \frac{114}{296} = \frac{57}{148} = 0.385135\ldots,$$
$$P(3,3) = \frac{5132}{13754} = \frac{2566}{6877} = 0.373127\ldots$$

and therefore $P(3,2) > P(3,3)$. Consequently (1) *— the clause as transcribed in Section 1 from tex line* 1463 *of* [1]*, read in this model — is false.*

Nothing in the comparison needs decimals. Cross-multiplying the two unreduced fractions,

$$114 \cdot 13754 = 1\,567\,956 \;>\; 1\,519\,072 = 5132 \cdot 296, \tag{6}$$

and the gap is

$$P(3,2) - P(3,3) = \frac{57}{148} - \frac{2566}{6877} = \frac{12221}{1017796} = 0.0120073\ldots, \tag{7}$$

about 1.2 percentage points. Here $n = 3$ is a fixed n and both $k = 2$ and $k = 3$ satisfy the conjecture's own "Fix $k \geq 2$", so the counterexample lies inside the clause's own quantifiers. The value $k = 1$ is *not* used: $P(3,1) = 1 > P(3,2)$ is also a decrease, but $k = 1$ is outside the source's range.

The rest of this section proves Theorem 1 in closed form, so that a reader needs no program. Throughout, $n = 3$ and indices are read modulo 3.

Step 1: total simplicity at $n = 3$ is three column inequalities.

Proposition 2. *A 3-vertex k-out graph with edge multiplicities a_{ij} is totally simple if and only if in each column the two off-diagonal entries differ:*

$$a_{10} \neq a_{20}, \qquad a_{01} \neq a_{21}, \qquad a_{02} \neq a_{12}.$$

Proof. By (3) the partitions to be excluded are those other than Δ and ∇, and the only such partitions of a 3-set are the three of shape $2 + 1$. Fix $\mathcal{C} = \{\{i, i'\}, \{j\}\}$. The singleton block imposes nothing. For the block $\{i, i'\}$, lumpability requires $|\delta_i(\{j\})| = |\delta_{i'}(\{j\})|$, i.e. $a_{ij} = a_{i'j}$, and $|\delta_i(\{i, i'\})| = |\delta_{i'}(\{i, i'\})|$; since each row sums to k, the latter reads $k - a_{ij} = k - a_{i'j}$ and is equivalent to the former. So $\mathcal{C}$ is lumpable exactly when the two off-diagonal entries of column j agree, and total simplicity is the conjunction of the three negations. $\square$

Step 2: the denominator $B(3, k)$.

Proposition 3. $B(3,k) = (3^k - 1)^3 - 3\,(2^k - 1)^2(3^k - 1)$*; in particular*

$$B(3,2) = 8^3 - 3 \cdot 3^2 \cdot 8 = 512 - 216 = 296,$$
$$B(3,3) = 26^3 - 3 \cdot 7^2 \cdot 26 = 17\,576 - 3822 = 13\,754,$$

and $B(3,4) = 80^3 - 3 \cdot 15^2 \cdot 80 = 512\,000 - 54\,000 = 458\,000.$

Proof. For a vertex v let $S_v = \{w \neq v : a_{vw} > 0\}$. A 3-vertex state fails to be strongly connected if and only if some S_v is empty (an *all-loop* vertex) or there is a *closed pair* $\{u, v\}$, meaning $S_u = \{v\}$ and $S_v = \{u\}$. Indeed, if every S_v is nonempty and the state is not strongly connected, take a sink component C of the condensation: every vertex of C has its out-support inside C, so $|C| = 1$ is impossible and $|C| = 3$ would be strong connectivity, leaving $|C| = 2$, which is a closed pair.

The ordered weight of the vertices v with $S_v \neq \emptyset$ is $3^k - 1$ each (all 3^k head choices, minus the single all-loop choice), giving $(3^k - 1)^3$ for the first term. For a fixed closed pair $\{u, v\}$, vertex u must send all k heads into $\{u, v\}$

with at least one to v, of weight $2^k - 1$, and likewise for v; the third vertex is unrestricted apart from $S_w \neq \emptyset$, of weight $3^k - 1$. The three closed pairs are pairwise incompatible, since two of them share a vertex, which would then be forced to be all-loops and hence to have $S = \emptyset$. Subtracting gives the formula. □

Step 3: the numerator $A(3,k)$ as a matrix trace. Give vertex i the ordered pair $(s_i, t_i) = (a_{i,i+1}, a_{i,i+2})$, so that $a_{ii} = k - s_i - t_i$ and the ordered weight of vertex i is the trinomial

$$W_k[s][t] = \frac{k!}{s!\,t!\,(k-s-t)!} \quad (s + t \leq k), \qquad W_k[s][t] = 0 \quad (s + t > k),$$

a symmetric $(k+1)\times(k+1)$ matrix. In these coordinates $a_{10} = t_1$, $a_{20} = s_2$, $a_{01} = s_0$, $a_{21} = t_2$, $a_{02} = t_0$, $a_{12} = s_1$, so Proposition 2 becomes the cyclic system

$$t_0 \neq s_1, \qquad t_1 \neq s_2, \qquad t_2 \neq s_0. \tag{8}$$

Let $D = J - I$ be the all-ones-minus-identity matrix of size $k + 1$ and put $f(t) = \sum_{q\geq 1} W_k[t][q]$.

Proposition 4. $A(3,k) = \operatorname{tr}((W_kD)^3) - 3\Big[\big(\sum_t f(t)\big)^2 - \sum_t f(t)^2\Big]$.

Proof. Since $(W_kD)[s][s'] = \sum_{t\neq s'} W_k[s][t]$, expanding the trace gives $\operatorname{tr}((W_kD)^3) = \sum \prod_i W_k[s_i][t_i]$ over all $(s_i,t_i)_{i=0,1,2}$ obeying (8): that is exactly the total ordered weight of the totally simple states, strong connectivity not yet imposed.

It remains to subtract the totally simple states that are not strongly connected. If a state is totally simple then, by the sink-component argument in the proof of Proposition 3 together with Proposition 2, it has no closed pair: a closed pair $\{u, v\}$ forces $a_{uw} = a_{vw} = 0$ in the column of the third vertex w. Hence a totally simple state fails strong connectivity precisely when some vertex is all-loops, and at most one vertex can be, since two all-loop vertices i, i' would give $a_{iw} = a_{i'w} = 0$. Say vertex 0 is all-loops, i.e. $s_0 = t_0 = 0$, of weight $W_k[0][0] = 1$. Then (8) reads $s_1 \geq 1$, $t_2 \geq 1$ and $t_1 \neq s_2$, while the requirement that vertices 1 and 2 are not all-loops is automatic. Summing over $s_1 \geq 1$ at fixed t_1 contributes $\sum_{s\geq 1} W_k[s][t_1] = f(t_1)$ by symmetry of W_k, and summing over $t_2 \geq 1$ at fixed s_2 contributes $f(s_2)$; the surviving constraint is $t_1 \neq s_2$. So the count is $\sum_{t_1\neq s_2} f(t_1)f(s_2) = (\sum f)^2 - \sum f^2$, and there are three choices of the all-loop vertex. □

Step 4: the arithmetic, in full. For $k = 2$ and $k = 3$ the matrices are, with rows indexed by s and columns by t,

$$W_2 = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 2 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \qquad W_3 = \begin{pmatrix} 1 & 3 & 3 & 1 \\ 3 & 6 & 3 & 0 \\ 3 & 3 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

with row sums $4, 4, 1$ summing to $3^2 = 9$ and $8, 12, 6, 1$ summing to $3^3 = 27$. Their traces and corrections are

$$\operatorname{tr}((W_2 D)^3) = 150, \qquad f = (3, 2, 0), \qquad 3[5^2 - 13] = 36,$$
$$\operatorname{tr}((W_3 D)^3) = 5798, \qquad f = (7, 9, 3, 0), \qquad 3[19^2 - 139] = 666,$$

so that $A(3, 2) = 150 - 36 = 114$ and $A(3, 3) = 5798 - 666 = 5132$. Together with Proposition 3 this proves Theorem 1: all four integers are obtained from a 3×3 and a 4×4 integer matrix and two subtractions, and (6) concludes. The same route at $k = 4$ gives $\operatorname{tr}((W_4 D)^3) = 189\,518$, $f = (15, 28, 18, 4, 0)$, correction $3[65^2 - 1349] = 8628$, hence $A(3, 4) = 180\,890$ and

$$P(3, 4) = \frac{180\,890}{458\,000} = \frac{18089}{45800} = 0.394956 \ldots > P(3, 3), \tag{9}$$

so monotonicity resumes at the very next step. The corrections 36, 666 and 8628 are exactly the weights of the states that are totally simple and *not* strongly connected.

Propositions 3 and 4 give $A(3, k)$ and $B(3, k)$ in closed form for every k. Evaluating them and comparing consecutive values by exact cross-multiplication is a finite computation, and the program of Section 6 carried it out for $2 \leq k \leq 30$: in that range $k = 2 \to 3$ is the only step along $n = 3$ at which (1) fails, and $P(3, k) < P(3, k + 1)$ strictly for $3 \leq k \leq 29$. This is a verified computation rather than a consequence of the closed forms alone, and it says nothing about $k > 30$.

4. Total simplicity does not imply strong connectivity

This is secondary to the counterexample; it is recorded because the numerator computation depends on it.

Proposition 5. *Some totally simple* 3*-vertex* 2*-out graphs are not strongly connected. Hence the sentence transcribed from tex line* 1455 *of* [1], *"if a k-out digraph G is not strongly connected, then it is automatically nontrivially lumpable", and the assertion of the source's remark at tex* 1516–1521 *that every totally simple k-out graph is strongly connected, are both false.*

Proof. The 2-out graph $0 \to \{0, 0\}$, $1 \to \{0, 2\}$, $2 \to \{1, 1\}$ on $\{0, 1, 2\}$, written as out-multisets, has $a_{10} = 1 \neq 0 = a_{20}$, $a_{01} = 0 \neq 2 = a_{21}$ and $a_{02} = 0 \neq 1 = a_{12}$, so it is totally simple by Proposition 2; yet vertex 0 is all loops and reaches nothing, so it is not strongly connected. The remark's argument takes a sink strongly connected component C and forms "the equivalence relation whose only non-singleton class is C"; when $|C| = 1$ that relation is Δ_G, which the source itself calls trivial at tex 355, and the graph just exhibited realises that gap. $\square$

Consequently $P(n, k)$ is *not* $\Pr(\mathrm{TS}) / \Pr(\mathrm{SCC})$: the joint weight $A(n, k)$ must be computed with the strong connectivity test applied, as in (4).

5. Status: what is settled, and what is not

The clause (1) is false as transcribed: one fixed n and one k-step inside its own quantifiers suffice, and Theorem 1 exhibits them with closed-form integers. Two things bound that. The clause, its quantifiers and the identification of the model are hand transcription from an e-print we cannot pin (Section 1); and the failure is in the ordered/with-replacement model only — the dip (7) vanishes under the rival reading of Remark 2, which records the rival numbers and the two features of the source that pin the reading used.

The two *asymptotic* clauses of the same conjecture — $P(n, k) \to 1$ as $n \to \infty$ (tex 1461) and the expansion $1 - \exp(-c_k n + \beta_k) + o(1)$ with c_k increasing (tex 1465–1470) — are not addressed here, and no finite census can address them. They remain open. The large-n intent of the conjecture survives everything computed here: the source's own sampled table at $n = 8, 9, 10$ (tex 1484–1506) is monotone in k at each of those three values of n, and $n = 3$ lies five below the smallest n it samples. Replacing "for fixed n" by "for fixed $n \geq 4$" is consistent with every cell we know of, but that is not a theorem: it is quantified over all $n \geq 4$ and all $k \geq 2$ and supported by finitely many cells. Nor do we claim that $k = 2 \to 3$ is the only violation along $n = 3$ beyond $k = 30$.

The denominators are not new. The numbers $B(n, 2)$ are the classical census of labelled strongly connected n-state 2-input automata, OEIS `A027834` [6], with $B(3, 2) = 296$ and $B(4, 2) = 20\,958$ the third and fourth terms; the sequence goes back to Harrison [2] and Liskovets [5]. Neither is $B(3, 3) = 13\,754$ new. An editorial note in OEIS `A006691` [7], quoting Robinson [4] and Liskovets [5], records `A006692` (49, 6877, 1854545, . . .) as a normalised count of labelled strongly connected automata with three inputs, the p-state value being $(p-1)!$ times its $(p-1)$st term; at $p = 3$ that is $2! \cdot 6877 = 13\,754$. On that reading $B(n, k)$ is on record for every k, not only for $k = 2$. We have not verified that normalisation independently, and the program checks only the two terms 296 and 20 958 against its own census.

Adjacent work settles nothing about the numerator or about k-monotonicity. Harrison's census of strongly connected sequential machines [3] and the subsequent literature on automorphism groups and quotients of strongly connected automata count or classify such automata without the lumpability refinement, and the sources just cited give $B(n, k)$ but not $A(n, k)$; none of them addresses the behaviour of $P(n, k)$ in k. We claim no priority for the lumpability-refined numerator $A(n, k)$: searches of `A027834`'s host database along both the k-axis $(114, 5132, 180890)$ and the n-axis $(1, 9, 114, 6720)$ return no sequence, which is a failed search and not evidence of novelty.

6. Reproduction and verification

The accompanying program `verify.py` (Python 3.9+, standard library only) independently re-derives the census integers of Theorem 1 and the other census values printed here, within the limits recorded below. It enumerates the state space exhaustively at $n = 3$ for $k = 1, 2, 3, 4$, at $n = 4$ for $k = 2$ and at $n = 2$ for $k = 2, 3$, deciding lumpability from the definition over *all* set partitions with Δ and ∇ removed; it re-runs $n = 3$, $k = 2, 3$ a second time as a direct enumeration over the 3^{kn} raw ordered edge tuples with weight 1 per tuple and no multinomial weight anywhere; it checks the two independent closed forms of Propositions 3 and 4 against the census and against each other for $2 \leq k \leq 30$; it verifies the total-weight identity n^{kn} of (2) in every cell and that Δ and ∇ are lumpable on every state; and it decides every inequality by exact integer or `Fraction` arithmetic. Its recorded run reports `VERDICT: ALL 56 CHECKS PASS`.

The limits of that program are recorded in its own output and repeated here. It does not recompute any cell with $n \geq 5$, nor $n = 4$ with $k \geq 3$, so the repair discussed in Section 5 is consistent with, not verified by, it; and it does not recompute the source's Monte Carlo table at $n = 8, 9, 10$, nor any attribution in Section 5 — of those it checks only that its own census reproduces the two hand-transcribed terms 296 and 20 958, and it does not touch the `A006691`/`A006692` normalisation. Above all, it does not read [1]: the quoted clause, the tex line numbers, the definitions of Section 2 and the identification of the probability model are transcribed from the e-print by hand, and the program checks arithmetic in the model as transcribed. Its output banner speaks of a counterexample to the clause and calls the clause refuted; that reading of the source is the part of this note a reader must check against [1] directly, and it is the residual risk recorded in Remark 2 and in Section 1.

References

[1] D. D'Angeli and E. Rodaro, *On totally synchronizing graphs*, arXiv:2607.17335v1, 19 July 2026. Abstract page arxiv.org/abs/2607.17335v1; e-print source arxiv.org/e-print/2607.17335v1. Line numbers quoted in this note refer to the LaTeX source file `totally_main.tex` of that e-print. We supply no copy or hash of that file.

[2] M. A. Harrison, *A census of finite automata*, Canad. J. Math. **17** (1965), 100–113.

[3] M. A. Harrison, *Enumeration of strongly connected sequential machines*, Inform. and Control **8** (1965), doi:10.1016/S0019-9958(65)90316-5.

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[6] OEIS Foundation Inc., *Sequence `A027834`: number of labeled strongly connected n-state 2-input automata*, The On-Line Encyclopedia of Integer Sequences, oeis.org/A027834.

[7] OEIS Foundation Inc., *Sequences `A006691` (with the editorial note of P. Hadjicostas, February 2021) and `A006692`: connected n-state finite automata with 3 inputs*, The On-Line Encyclopedia of Integer Sequences, oeis.org/A006691, oeis.org/A006692.